Lecture Notes

Lecture 3

Linear Maps: Kernel, Image, Rank–Nullity

We have the objects of linear algebra—vector spaces—and their coordinate systems—bases. Now we study the *maps between* vector spaces that respect this structure. A *linear map* is a function that commutes with addition and scalar multiplication; equivalently, it preserves linear combinations, so it preserves superposition. This is exactly why linear maps are the right notion of "transformation" in physics: an observable, a symmetry, and the time-evolution of a quantum state are all linear operators, and the linearity is what makes the superposition principle survive the transformation.

Two subspaces measure how a linear map behaves: its *kernel* (what it sends to zero) and its *image* (what it can produce). The central result of the lecture, the rank–nullity theorem, ties their dimensions together in a single equation $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$, and from it we will read off when linear systems are solvable and when operators are invertible.

Learning objectives

By the end of this lecture you will be able to:

- Check linearity and build a map freely from its values on a basis.
- Compute the kernel and image and read off injectivity and surjectivity.
- Apply the rank–nullity theorem and its dimension obstructions.
- Recognise the quotient space and First Isomorphism Theorem (self-study).

1 Linear maps and their examples

Definition 3.1 (Linear map). Let V, W be vector spaces over the same field $\mathbb{F}$. A function $T\colon V \to W$ is a *linear map* (or *linear transformation*) if

$$T(u+v) = Tu + Tv \quad \text{and} \quad T(av) = aTv \qquad \text{for all } u, v \in V,\ a \in \mathbb{F}.$$

The set of all linear maps $V \to W$ is written $\mathcal{L}(V,W)$. A linear map from V to itself is an *operator*, and we write $\mathcal{L}(V)$ for $\mathcal{L}(V,V)$.

Applying the two rules repeatedly, a linear map sends 0 to 0 (take $a = 0$) and preserves every linear combination:

$$T(a_1v_1 + \cdots + a_mv_m) = a_1\,Tv_1 + \cdots + a_m\,Tv_m. \tag{3.1}$$

This single identity is the reason linear maps are ubiquitous.

Example 3.2 (A catalogue of linear maps). Each of the following is linear, as the two rules readily confirm.

- The *zero map* $0\colon V \to W$ and the *identity* $I\colon V \to V$, $Iv = v$.
- *Matrix maps*: for $A \in M_{m\times n}(\mathbb{F})$, the map $x \mapsto Ax$ from $\mathbb{F}^n$ to $\mathbb{F}^m$. (We will see in Lecture 4 that, in coordinates, *every* linear map looks like this.)
- *Differentiation* $D\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, $Dp = p'$, and more generally on spaces of differentiable functions.
- *Multiplication* $M\colon \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$, $(Mp)(t) = t\,p(t)$.

- The *backward shift* on $\mathbb{F}^\infty$, $T(x_1, x_2, x_3, \dots) = (x_2, x_3, \dots)$.
- *Rotation* of $\mathbb{R}^2$ by an angle θ, and the *evaluation* functional $\mathcal{P}(\mathbb{F}) \to \mathbb{F}$, $p \mapsto p(c)$. ◁

Example 3.3 (Testing linearity). The map $S(x, y) = (2x - y,\ x)$ on $\mathbb{R}^2$ is linear: each output coordinate is a linear combination of x, y with no constant term. The map $T(x, y) = (x+1,\ y)$ is *not* linear—it fails $T0 = 0$, since $T(0,0) = (1,0) \neq 0$. A constant term anywhere is fatal to linearity. ◁

Example 3.4 (Why the catalogue works). Differentiation is linear because of the single identity $(af+bg)' = af'+bg'$. The same "apply the rule to each piece" pattern—valid for the backward shift, integration, multiplication by a fixed function, and matrix multiplication—is what makes every entry of Example 3.2 linear. Linearity is rarely hard to check; the hard question is what a given linear map *does*, which is the rest of the lecture. ◁

2 A linear map is determined by a basis

A linear map cannot be chosen freely at every point: (3.1) ties its values together. In fact its values on a single basis fix it completely, and those values may be assigned arbitrarily.

> **Theorem 3.5 (Maps are determined by a basis).** Let $v_1, \dots, v_n$ be a basis of V and let $w_1, \dots, w_n$ be *any* vectors in W. Then there is a unique linear map $T\colon V \to W$ with $Tv_k = w_k$ for each k.

Proof. Existence. Each $v \in V$ has a unique expression $v = \sum_k a_k v_k$ (Lecture 2); define $Tv = \sum_k a_k w_k$. Linearity follows because coordinates add and scale: if $v = \sum a_k v_k$ and $u = \sum b_k v_k$ then $v + u = \sum (a_k + b_k) v_k$, so $T(v+u) = \sum (a_k + b_k) w_k = Tv + Tu$, and similarly $T(cv) = c\,Tv$. Clearly $Tv_k = w_k$.

Uniqueness. If S is linear with $Sv_k = w_k$, then for any $v = \sum a_k v_k$, (3.1) gives $Sv = \sum a_k w_k = Tv$. So $S = T$. ∎

> **Corollary 3.6 (Agreement on a basis).** If two linear maps $S, T \in \mathcal{L}(V, W)$ satisfy $Sv_k = Tv_k$ on a basis $v_1, \dots, v_n$ of V, then $S = T$.

Proof. Both are linear maps with the same values on the basis, so both equal the unique map of Theorem 3.5. ∎

Example 3.7 (Building a map from its values on a basis). Define $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ by its action on the basis $(1,1), (1,-1)$:

$$T(1,1) = (2,0), \qquad T(1,-1) = (0,2).$$

To evaluate T on a general (x, y), first find coordinates: writing $(x, y) = a(1,1) + b(1,-1)$ gives $a = \frac{x+y}{2}$, $b = \frac{x-y}{2}$. Then

$$T(x, y) = a\,(2,0) + b\,(0,2) = (2a,\ 2b) = (x + y,\ x - y).$$

Theorem 3.5 guarantees this is the one and only linear map with the prescribed values. ◁

3 The algebra of linear maps

Linear maps can be added, scaled, and composed.

Definition 3.8 (Operations on linear maps). For $S, T \in \mathcal{L}(V, W)$ and $a \in \mathbb{F}$, define $S + T$ and aS pointwise: $(S + T)v = Sv + Tv$ and $(aS)v = a(Sv)$. For $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, the *product* (composition) $ST \in \mathcal{L}(U, W)$ is $(ST)u = S(Tu)$.

Proposition 3.9 (The algebra of linear maps). With the pointwise operations, $\mathcal{L}(V, W)$ is a vector space. The composition of linear maps is linear, and composition is associative, distributes over addition, and has the identity as a unit: $(RS)T = R(ST)$, $R(S + T) = RS + RT$, $(R + S)T = RT + ST$, and $IT = TI = T$.

Proof. $\mathcal{L}(V, W)$ is a subspace of the space W^V of all functions $V \to W$: the zero map is linear, and if S, T are linear and $a \in \mathbb{F}$ then so are $S + T$ and aS, since e.g. $(S + T)(u + v) = S(u+v)+T(u+v) = Su+Sv+Tu+Tv = (S+T)u+(S+T)v$. That ST is linear follows by applying T and then S. The remaining identities are the ordinary laws of function composition combined with linearity, verified by evaluating each side at an arbitrary vector. ■

Products, however, **need not commute**—and the failure to commute carries physical content.

Example 3.10 (A commutator: the canonical commutation relation). On $\mathcal{P}(\mathbb{R})$ let D be differentiation and M multiplication by t. For any p,

$$(DM)p = \big(t\,p(t)\big)' = p(t) + t\,p'(t), \qquad (MD)p = t\,p'(t),$$

so $(DM - MD)p = p$ for every p; that is, the *commutator* $[D, M] := DM - MD$ equals the identity:

$$[D, M] = I. \tag{3.2}$$

In quantum mechanics the position and momentum operators obey the analogous relation $[\hat{x}, \hat{p}] = i\hbar I$. Non-commuting operators are the rule, not the exception, and (3.2) is your first sighting of why. ◁

Example 3.11 (Concrete non-commuting operators). The phenomenon is visible already with 2×2 matrices. Let $S, T \in \mathcal{L}(\mathbb{R}^2)$ be the matrix maps with $S = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $T = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$. Then

$$ST = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad TS = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad [S, T] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \neq 0.$$

(S and T are the raising and lowering operators of a two-level system.) ◁

Remark 3.12. Once bases of V and W are fixed, addition, scaling, and composition of linear maps become exactly addition, scaling, and multiplication of their matrices—the subject of Lecture 4. One can show $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$; the proof is cleanest in matrix language, so we defer it.

4 Kernel and injectivity

Definition 3.13 (Null space / kernel). The *null space* (or *kernel*) of $T \in \mathcal{L}(V, W)$ is

$$\operatorname{null} T = \{v \in V : Tv = 0\} \subseteq V.$$

Proposition 3.14 (The kernel is a subspace). null T is a subspace of V.

Proof. $T0 = 0$, so $0 \in \operatorname{null} T$. If $u, v \in \operatorname{null} T$ and $a \in \mathbb{F}$ then $T(u+v) = Tu + Tv = 0$ and $T(av) = aTv = 0$, so $u + v$ and av lie in null T. The subspace test is met. ■

Example 3.15 (Kernels). The kernel of differentiation D on $\mathcal{P}(\mathbb{R})$ is the constants (the functions with zero derivative). The kernel of multiplication by t is $\{0\}$. The kernel of the backward shift on $\mathbb{F}^\infty$ is $\{(a, 0, 0, \dots) : a \in \mathbb{F}\}$. And the kernel of a matrix map $x \mapsto Ax$ is exactly the solution set of the homogeneous system $Ax = 0$—the null space you met when row-reducing A. ◁

Example 3.16 (Computing a kernel). Let $T: \mathbb{R}^3 \to \mathbb{R}^2$ be $Tx = Ax$ with $A = \left(\begin{smallmatrix} 1 & 2 & -1 \\ 2 & 4 & 1 \end{smallmatrix}\right)$. The kernel is the solution set of $Ax = 0$. Subtracting twice the first row from the second gives $(0, 0, 3)$, so $x_3 = 0$; the first row then reads $x_1 + 2x_2 = 0$, i.e. $x_1 = -2x_2$, with x_2 free. Hence

$$\operatorname{null} T = \operatorname{span}\big((-2, 1, 0)\big), \qquad \dim \operatorname{null} T = 1.$$

We return to this map in Example 3.22 to see its range. ◁

Definition 3.17 (Injective). $T: V \to W$ is *injective* (one-to-one) if $Tu = Tv$ implies $u = v$.

Theorem 3.18 (Injectivity is a kernel condition). $T \in \mathcal{L}(V, W)$ is injective if and only if $\operatorname{null} T = \{0\}$.

Proof. If T is injective and $v \in \operatorname{null} T$, then $Tv = 0 = T0$ forces $v = 0$, so $\operatorname{null} T = \{0\}$. Conversely, if $\operatorname{null} T = \{0\}$ and $Tu = Tv$, then $T(u-v) = Tu - Tv = 0$, so $u - v \in \operatorname{null} T = \{0\}$, giving $u = v$. Thus T is injective. ■

5 Image and surjectivity

Definition 3.19 (Range / image; rank). The *range* (or *image*) of $T \in \mathcal{L}(V, W)$ is $\operatorname{range} T = \{Tv : v \in V\} \subseteq W$, and its dimension $\dim \operatorname{range} T$ is the *rank* of T. The map T is *surjective* (onto) if $\operatorname{range} T = W$.

Proposition 3.20 (The image is a subspace). range T is a subspace of W.

Proof. $0 = T0 \in \operatorname{range} T$. If $w_1 = Tv_1$ and $w_2 = Tv_2$ then $w_1 + w_2 = T(v_1 + v_2) \in \operatorname{range} T$, and $aw_1 = T(av_1) \in \operatorname{range} T$. The subspace test holds. ■

Example 3.21 (Ranges, and dependence on the target). The range of a matrix map $x \mapsto Ax$ is the span of the columns of A—the column space. Differentiation $D: \mathcal{P}(\mathbb{R}) \to \mathcal{P}(\mathbb{R})$ is surjective, since every polynomial has an antiderivative. But surjectivity depends on the chosen target: $D: \mathcal{P}_5(\mathbb{R}) \to \mathcal{P}_5(\mathbb{R})$ is *not* surjective (t^5 is not a derivative of anything in $\mathcal{P}_5$), whereas $D: \mathcal{P}_5(\mathbb{R}) \to \mathcal{P}_4(\mathbb{R})$ *is*. ◁

Example 3.22 (Computing a range). For the map $T: \mathbb{R}^3 \to \mathbb{R}^2$ of Example 3.16, the range is the column space of A. Its columns are $(1, 2), (2, 4), (-1, 1)$; since $(2, 4) = 2(1, 2)$, two independent columns remain, so

$$\operatorname{range} T = \operatorname{span}\big((1, 2), (-1, 1)\big) = \mathbb{R}^2, \qquad \operatorname{rank} T = 2.$$

Thus T is surjective, and the dimensions balance: $3 = \dim \operatorname{null} T + \operatorname{rank} T = 1 + 2$, a preview of the theorem below. ◁

A linear map carries spanning lists to spanning lists, and—when injective—it preserves independence.

Proposition 3.23 (Action on lists). Let $T \in \mathcal{L}(V, W)$ and $v_1, \ldots, v_n \in V$.

1. If $v_1, \ldots, v_n$ spans V, then $Tv_1, \ldots, Tv_n$ spans range T.
2. If T is injective and $v_1, \ldots, v_n$ is linearly independent, then $Tv_1, \ldots, Tv_n$ is linearly independent.

Proof. (1) Every element of range T is Tv for some $v = \sum_k a_k v_k$, and then $Tv = \sum_k a_k Tv_k$ lies in $\operatorname{span}(Tv_1, \ldots, Tv_n)$. (2) If $\sum_k c_k Tv_k = 0$ then $T\left(\sum_k c_k v_k\right) = 0$, so $\sum_k c_k v_k$ is in $\operatorname{null} T = \{0\}$ by injectivity; independence of the v_k then forces every $c_k = 0$. ■

6 The Fundamental Theorem of Linear Maps

The kernel records what T destroys; the range records what it builds. The next theorem—the central result of the lecture—says their dimensions exactly account for the dimension of the domain (Figure 1).

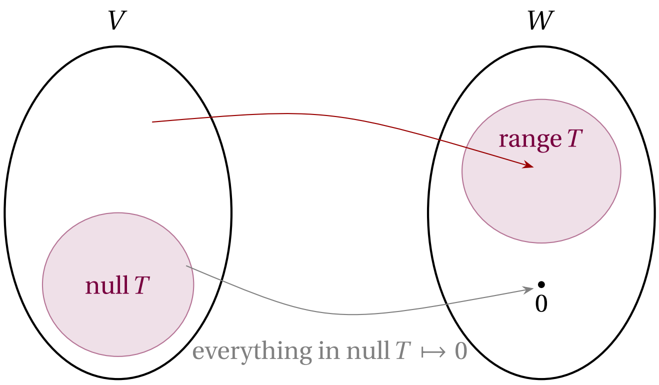

Figure 1: A linear map collapses its kernel to 0 and carries a complement of the kernel one-to-one onto its image; the dimensions balance as $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$.

Theorem 3.24 (Rank–nullity). Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then range T is finite-dimensional and

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T.$$

Proof. Let $u_1, \ldots, u_m$ be a basis of null T, so $\dim \operatorname{null} T = m$. Extend it (Lecture 2) to a basis $u_1, \ldots, u_m, v_1, \ldots, v_n$ of V, so $\dim V = m + n$. We show $Tv_1, \ldots, Tv_n$ is a basis of range T, which gives $\dim \operatorname{range} T = n$ and finishes the count.

Spanning. Any $v \in V$ is $v = \sum_i a_i u_i + \sum_j b_j v_j$; applying T and using $Tu_i = 0$ leaves $Tv = \sum_j b_j Tv_j$. Hence $Tv_1, \ldots, Tv_n$ spans range T (and range T is finite-dimensional).

Independence. Suppose $\sum_j c_j Tv_j = 0$. Then $T\left(\sum_j c_j v_j\right) = 0$, so $\sum_j c_j v_j \in \operatorname{null} T$ and can be written $\sum_i d_i u_i$. Thus $\sum_j c_j v_j - \sum_i d_i u_i = 0$; independence of the basis $u_1, \ldots, u_m, v_1, \ldots, v_n$ forces all $c_j = 0$ (and all $d_i = 0$). So $Tv_1, \ldots, Tv_n$ is independent, hence a basis of range T. ■

Example 3.25 (A projection makes the count visible). Let $P\colon \mathbb{R}^2 \to \mathbb{R}^2$ be projection onto the x-axis, $P(x,y) = (x,0)$ (Figure 2). Its kernel is the y-axis (mapped to 0) and its range is the x-axis, so $\dim \operatorname{null} P = 1$ and $\dim \operatorname{range} P = 1$, balancing $2 = 1+1 = \dim \mathbb{R}^2$. ◁

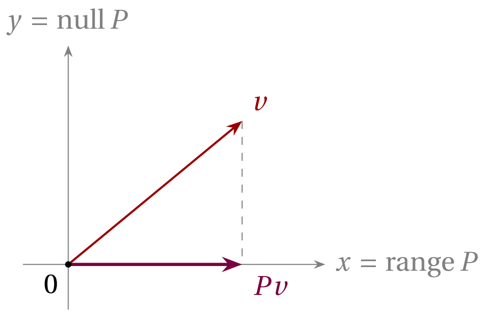

Figure 2: Projection onto the x-axis: the kernel is the y-axis, the range is the x-axis, and $\dim \operatorname{null} P + \dim \operatorname{range} P = 1 + 1 = 2$.

Two consequences follow with no further work, just by comparing dimensions.

Corollary 3.26 (Dimension obstructions). Let V, W be finite-dimensional and $T \in \mathcal{L}(V,W)$.

1. If $\dim V > \dim W$, then T is not injective.
2. If $\dim V < \dim W$, then T is not surjective.

Proof. (1) $\dim \operatorname{null} T = \dim V - \dim \operatorname{range} T \geq \dim V - \dim W > 0$, so $\operatorname{null} T \neq \{0\}$ and T is not injective by Theorem 3.18. (2) $\dim \operatorname{range} T = \dim V - \dim \operatorname{null} T \leq \dim V < \dim W$, so $\operatorname{range} T \neq W$. ■

Example 3.27 (Reading injectivity and surjectivity off dimensions). No computation is needed to see that $T(z_1, z_2, z_3, z_4) = (z_1 + z_4,\ z_2 + z_3,\ z_1 + z_2)$ from $\mathbb{R}^4$ to $\mathbb{R}^3$ has a nonzero kernel: since $\dim \mathbb{R}^4 > \dim \mathbb{R}^3$, Corollary 3.26(1) forbids injectivity. Dually, no linear map $\mathbb{R}^2 \to \mathbb{R}^3$ is surjective, so for any such map the system $Tx = b$ is unsolvable for some right-hand sides b. Dimension alone settles both questions. ◁

Corollary 3.28 (Operators: injective = surjective). If V is finite-dimensional and $T \in \mathcal{L}(V)$, then T is injective if and only if it is surjective (and then bijective).

Proof. By Theorem 3.24, $\dim \operatorname{null} T + \dim \operatorname{range} T = \dim V$. So $\operatorname{null} T = \{0\}$ (injective) $\iff \dim \operatorname{range} T = \dim V \iff \operatorname{range} T = V$ (surjective), using Theorem 3.18 and that a subspace of full dimension is the whole space. ■

Corollary 3.28 fails in infinite dimensions: the backward shift on $\mathbb{F}^\infty$ is surjective but not injective, while the forward shift is injective but not surjective. This contrast is a first warning of the subtleties that Lectures 13–14 confront in Hilbert space.

Example 3.29 (A rotation is an invertible operator). Rotation of $\mathbb{R}^2$ by an angle θ,

$$R_\theta(x,y) = (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta),$$

is linear, and its kernel is $\{0\}$ (a rotation carries every nonzero vector to a nonzero vector). By Corollary 3.28 it is therefore surjective, hence invertible—its inverse is $R_{-\theta}$, as geometry demands. Symmetry operations in physics are invertible linear operators for precisely this reason. ◁

Example 3.30 (Solving an equation by counting dimensions). Given $q \in \mathcal{P}_m(\mathbb{R})$, does some p satisfy $\big((t^2+1)p\big)'' = q$? Define $T : \mathcal{P}_m(\mathbb{R}) \to \mathcal{P}_m(\mathbb{R})$ by $Tp = \big((t^2+1)p\big)''$: multiplying by t^2+1 raises the degree by 2 and two derivatives lower it by 2, so T does map $\mathcal{P}_m$ into itself. If $Tp = 0$ then $(t^2+1)p$ has vanishing second derivative, hence is affine; but $(t^2+1)p$ has degree at least 2 unless $p = 0$. So $\operatorname{null} T = \{0\}$, i.e. T is injective, and by Corollary 3.28 it is surjective. Thus the required p exists for *every* q—injectivity delivered existence with no construction at all. ◁

Linear systems, revisited

Rank–nullity is the structural statement behind the row-reduction you already know. For $A \in M_{m\times n}(\mathbb{F})$ and $T(x) = Ax$, the rank of T is the number of pivots and the nullity is the number of free variables, so

$$(\#\text{free variables}) = n - (\#\text{pivots}) = \dim V - \operatorname{rank} T.$$

The system $Ax = b$ is solvable exactly when $b \in \operatorname{range} T$ (the column space), and then its full solution set is a single particular solution plus the kernel, $x_p + \operatorname{null} T$ (a translate of a subspace). In particular, by Corollary 3.26(1), a homogeneous system with more unknowns than equations always has a nonzero solution.

Example 3.31 (A homogeneous system with free variables). The system $x_1 + x_2 - x_3 + 2x_4 = 0$, $2x_1 + 2x_2 + x_3 + x_4 = 0$ has four unknowns and only two equations, so it must have nonzero solutions. Subtracting twice the first equation from the second gives $3x_3 - 3x_4 = 0$, i.e. $x_3 = x_4$; the first then gives $x_1 = -x_2 - x_4$, with x_2, x_4 free. The solution space is

$$\operatorname{span}\big((-1,1,0,0),\ (-1,0,1,1)\big), \qquad \text{of dimension } 2 = 4-2.$$

◁

Example 3.32 (Rank, nullity, and a solution set). Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be $Tx = Ax$ with

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{pmatrix}.$$

Row reduction gives pivot rows $(1,2,0,1)$ and $(0,0,1,1)$ (the third row is their consequence), so $\operatorname{rank} T = 2$ and, by Theorem 3.24, $\dim \operatorname{null} T = 4-2 = 2$. Solving $x_1 = -2x_2 - x_4,\ x_3 = -x_4$ with x_2, x_4 free,

$$\operatorname{null} T = \operatorname{span}\big((-2,1,0,0),\ (-1,0,-1,1)\big).$$

The range is the column space $\operatorname{span}\big((1,2,1),(0,1,1)\big)$, also of dimension 2, confirming $4 = 2+2$. Finally $b = (1,3,2)$ is the fourth column Ae_4, so $x_p = (0,0,0,1)$ solves $Ax = b$ and the complete solution set is $x_p + \operatorname{null} T$. ◁

Example 3.33 (Rank–nullity without matrices). On $\mathcal{P}_3(\mathbb{R})$, which has dimension 4, let D be differentiation. Its kernel is the constants, so $\dim \operatorname{null} D = 1$; its range is all derivatives of cubics, namely $\mathcal{P}_2(\mathbb{R})$, of dimension 3. The theorem checks out: $4 = 1+3$. The same D is neither injective nor surjective as an operator on $\mathcal{P}_3(\mathbb{R})$, yet becomes surjective when viewed as a map $\mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$: rank and nullity are intrinsic to the map, but injectivity and surjectivity depend on the chosen domain and target. ◁

Remark 3.34 (Operators in quantum mechanics). Every transformation a physicist applies to a quantum state—a symmetry, a measurement's observable, the time-evolution operator $e^{-i\hat{H}t/\hbar}$—is linear, precisely so that superpositions transform coherently: $T(a|\psi\rangle + b|\phi\rangle) = a\,T|\psi\rangle + b\,T|\phi\rangle$. Kernels and ranges recur throughout: the kernel of $\hat{H} - EI$ is the eigenspace at energy E, the object the spectral theorem (Lecture 10) will put at centre stage.

Quotient spaces *[self-study]*

The solution set $x_p + \operatorname{null} T$ of $Ax = b$ is not a subspace—it does not contain 0 if $x_p \neq 0$—yet it has a clear structure: it is a *translate* of the subspace $\operatorname{null} T$. The right language for objects like this is the quotient space.

Definition 3.35 (Quotient space). Let U be a subspace of V. The *coset* of $v \in V$ modulo U is

$$v + U = \{v + u : u \in U\}.$$

Two vectors are *equivalent*, $v \sim v'$, if $v - v' \in U$. The *quotient space* V/U is the set of all cosets, with operations

$$(v + U) + (w + U) = (v + w) + U, \qquad \lambda(v + U) = (\lambda v) + U,$$

making it a vector space of dimension $\dim V - \dim U$.

The coset $x_p + \operatorname{null} T$ from the previous subsection is *exactly* an element of $V/\operatorname{null} T$: the solution set of $Ax = b$ is a single point in the quotient space, not in V itself. Different right-hand sides b correspond to different cosets—the quotient "forgets" what happens inside the kernel.

Example 3.36 (Cosets in $\mathbb{R}^3$). Let $U = \operatorname{span}\big((1,0,0)\big) \subseteq \mathbb{R}^3$ (the x-axis). The coset of (a, b, c) is the line $(a, b, c) + U = \{(a + t, b, c) : t \in \mathbb{R}\}$, a horizontal line through (a, b, c). Cosets with the same y- and z-coordinates are equal, so $\mathbb{R}^3/U \cong \mathbb{R}^2$ (the yz-plane), of dimension $3 - 1 = 2$. ◁

Theorem 3.37 (First Isomorphism Theorem). For $T \in \mathcal{L}(V, W)$, the map $\tilde{T} : V/\operatorname{null} T \to W$ defined by

$$\tilde{T}(v + \operatorname{null} T) = Tv$$

is well-defined, linear, and injective, with $\operatorname{range} \tilde{T} = \operatorname{range} T$. Hence

$$V/\operatorname{null} T \cong \operatorname{range} T,$$

which (using $\dim(V/\operatorname{null} T) = \dim V - \dim \operatorname{null} T$) recovers the rank–nullity equation $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$ by comparing dimensions—an *alternative* view of Theorem 3.24, already proved directly above, rather than an independent proof of it.

Proof. *Well-defined.* If $v + \operatorname{null} T = v' + \operatorname{null} T$ then $v - v' \in \operatorname{null} T$, so $T(v - v') = 0$ and $Tv = Tv'$.

Linear. $\tilde{T}\big((v+\operatorname{null} T)+(w+\operatorname{null} T)\big) = T(v+w) = Tv+Tw = \tilde{T}(v+\operatorname{null} T)+\tilde{T}(w+\operatorname{null} T)$, and similarly for scalars.

Injective. $\tilde{T}(v + \operatorname{null} T) = 0$ means $Tv = 0$, i.e. $v \in \operatorname{null} T$, so $v + \operatorname{null} T = \operatorname{null} T$ is the zero element of $V/\operatorname{null} T$.

The range of $\tilde{T}$ is the range of T, and the dimension equation is $\dim(V/\operatorname{null} T)+\dim \operatorname{null} T = (\dim V - \dim \operatorname{null} T) + \dim \operatorname{null} T = \dim V$. ■

The First Isomorphism Theorem is the canonical form of rank–nullity: it does not merely count dimensions but exhibits an explicit isomorphism.

Remark 3.38 (Physics: gauge fields and cohomology *[self-study]*). Quotient spaces appear at the foundations of modern physics.

Gauge symmetry. In electromagnetism the 4-potential A_μ is not unique: two potentials related by $A_\mu \to A_\mu + \partial_\mu \chi$ produce the same field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The "pure gauge" configurations (those of the form $\partial_\mu \chi$) form a subspace of the space of all potentials, and the space of physically distinct fields is the quotient. A gauge-fixed description chooses one representative from each coset—just as choosing x_p picks one solution from each coset $x_p + \ker T$.

De Rham cohomology. For differential forms, the exterior derivative d satisfies $d^2 = 0$ (the analogue of a nilpotent, Lecture 6). The *closed* forms $Z^k = \ker d^k$ and the *exact* forms $B^k = \operatorname{range} d^{k-1}$ satisfy $B^k \subseteq Z^k$, so the de Rham cohomology $H^k = Z^k / B^k$ is a quotient of vector spaces. Its dimension records the topology of the manifold and governs the Aharonov–Bohm effect, magnetic monopoles, and Chern numbers.

BRST cohomology. In gauge quantum field theories the physical Hilbert space is $\ker Q / \operatorname{range} Q$, where Q is the nilpotent BRST charge ($Q^2 = 0$). Physical states are equivalence classes: two states are identified if they differ by Q-exact terms. This quotient space is what guarantees unitarity and the decoupling of unphysical degrees of freedom.

Summary of main concepts

A linear map preserves linear combinations, hence superposition. It is fixed by its values on a basis, which may be chosen freely (Theorem 3.5). Its kernel $\operatorname{null} T$ and image $\operatorname{range} T$ are subspaces; T is injective exactly when $\operatorname{null} T = \{0\}$ and surjective exactly when $\operatorname{range} T = W$. The rank–nullity theorem $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$ ties the two together, yielding at a glance that maps to smaller spaces cannot be injective, maps to larger spaces cannot be surjective, and operators on a finite-dimensional space are injective if and only if surjective. In coordinates this is the bookkeeping of pivots and free variables behind every linear system.

- **Linear map**—preserves linear combinations (superposition); fixed by its values on a basis.
- **Kernel and image**—subspaces $\operatorname{null} T$ and $\operatorname{range} T$; injective $\iff$ $\operatorname{null} T = \{0\}$, surjective $\iff$ $\operatorname{range} T = W$.
- **Rank–nullity**—$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$.
- **Operator dichotomy**—on a finite-dimensional space, injective $\iff$ surjective.
- **Quotient space; First Isomorphism Theorem**—$V / \operatorname{null} T \cong \operatorname{range} T$ (self-study); gauge equivalence in physics.

Exercises for this lecture are in the companion sheet `exercises-3.pdf`.